MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 117465203
MEMORY I

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Power Control.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

117465203	Circuit Card Assembly, Memory I

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item identification.- This assembly is one of two single ended printed circuit board consisting 512 bits, Bi-Polar programmable read - only - memories which supply the digital information for 100 programed flights for the Simulated Target Generator.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11745203.

3.2 Characteristics.

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3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with a + 5VDC Power Supply.

3.2.1.2 Input/Output Test.- When specific logic inputs are applied a corresponding output shall result as in Table I.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-G-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11745203 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-G-48411, Drawing 11745203 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-G-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-G-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-G-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-G-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-G-48411.

4.2.1 Test equipment.- (see Figure 1)

4.2.1.1 Power Supply.- The power supply shall be 10.00 $\pm$ 0.05 VDC, 100 mA or greater, an a ripple of less than 10 MV RMS

4.2.1.2 Dummy Loads.- The dummy load shall be eight triple 3-input positive nand gates type SN5410 or equilvalent.

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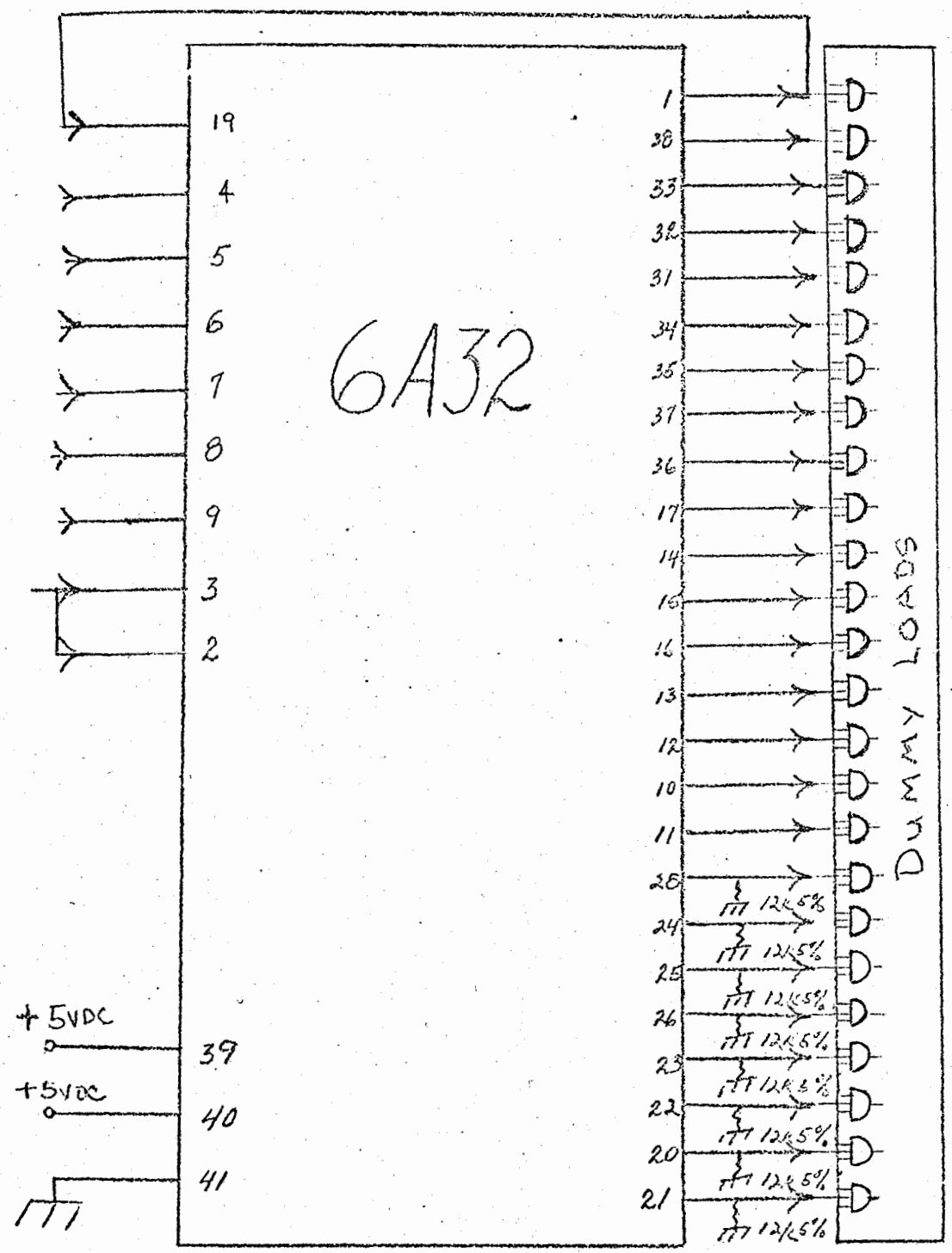


FIGURE 1

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TABLE I - INPUT/OUTPUT MEMORY I TEST

	INPUT							OUTPUT																						
Dec.	3	9	8	7	6	5	4	36	37	35	34	31	32	33	38	11	10	12	13	16	15	14	17	21	20	22	23	26	25	24	28
0	0	0	0	0	0	0	0	0	1	1	1	0	1	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	1	0	1	1	0	1	0	0	1	1	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	1	0	0	1	0	0	1	0	0	0	1	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	1	1	0	0	1	1	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0
4	0	0	0	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0
5	0	0	0	0	1	0	1	1	0	0	1	1	0	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	1	1	0	1	0	0	0	0	1	1	0	1	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	1	1	1	0	1	0	1	1	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
8	0	0	0	1	0	0	0	0	1	0	0	0	1	0	1	1	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0
9	0	0	0	1	0	0	1	1	0	0	1	0	1	0	0	0	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0
10	0	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0
11	0	0	0	1	0	1	1	1	0	0	1	1	0	0	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0
12	0	0	0	1	1	0	0	1	0	1	0	0	1	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0
13	0	0	0	1	1	0	1	0	0	0	0	1	0	0	1	1	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0
14	0	0	0	1	1	1	0	1	0	1	0	0	1	1	0	1	0	1	0	0	1	0	0	0	0	1	0	1	0	0	0
15	0	0	0	1	1	1	1	1	0	1	0	0	1	0	1	1	0	1	0	0	1	0	0	0	0	1	0	1	0	0	0
16	0	0	1	0	0	0	0	1	0	0	0	0	1	1	0	1	0	1	1	0	1	0	0	0	0	1	0	1	0	0	0
17	0	0	1	0	0	0	1	0	0	1	1	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0
18	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	1	1	0	1	0	0	0	0	0	0	1	0	0	0

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TABLE I - INPUT/OUTPUT MEMORY I TEST

	INPUT							OUTPUT																					
Dec.	3	9	8	7	6	5	4	36	37	35	34	31	32	33	38	11	10	12	13	16	15	14	17	21	20	22	23	26	25	24	28
19	0	0	1	0	0	1	1	1	0	0	1	0	0	0	1	0	1	1	1	0	0	1	0	1	0	0	0	0	1	0	0
20	0	0	1	0	1	0	0	1	1	0	0	1	1	0	0	1	0	0	1	0	0	1	0	1	1	0	0	0	1	1	0
21	0	0	1	0	1	0	1	1	1	0	1	0	1	0	1	0	1	1	1	0	1	0	0	1	0	0	0	0	1	1	0
22	0	0	1	0	1	1	0	1	0	0	1	0	0	0	1	0	1	1	1	0	0	1	0	1	0	0	0	0	1	0	0
23	0	0	1	0	1	1	1	1	1	0	1	0	1	0	0	0	1	1	1	0	1	0	0	1	0	0	0	0	1	1	0
24	0	0	1	1	0	0	0	0	1	0	0	1	0	0	0	1	0	1	0	0	0	1	0	1	1	0	0	0	1	1	0
25	0	0	1	1	0	0	1	0	0	1	1	0	0	0	0	0	1	0	0	0	0	1	0	1	0	0	0	0	1	1	1
26	0	0	1	1	0	1	0	0	0	0	1	1	1	0	1	0	0	0	0	0	1	0	0	1	1	0	0	0	1	1	0
27	0	0	1	1	0	1	1	1	1	1	0	1	0	0	0	1	0	0	0	0	1	0	0	1	1	0	0	0	1	0	0
28	0	0	1	1	1	0	0	1	0	1	0	1	1	0	0	1	0	0	0	0	1	0	1	0	0	0	0	0	0	1	0
29	0	0	1	1	1	0	1	1	0	1	0	1	1	0	0	1	0	0	0	0	1	0	0	1	1	0	0	0	0	1	0
30	0	0	1	1	1	1	0	0	1	0	0	0	1	0	1	1	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0
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35	0	1	0	0	0	1	1	0	0	0	0	0	1	1	0	1	1	0	1	0	1	1	0	0	0	1	0	1	0	0	0
36	0	1	0	0	1	0	0	1	0	0	0	0	1	0	1	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0
37	0	1	0	0	1	0	1	1	0	0	0	0	1	0	1	1	0	1	1	1	0	0	0	0	0	1	1	1	0	0	0
38	0	1	0	0	1	1	0	1	0	1	1	1	1	1	0	0	1	0	1	1	0	0	0	0	0	1	1	1	0	0	0

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TABLE I - INPUT/OUTPUT MEMORY I TEST

Dec.	INPUT							OUTPUT																						
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42	0	1	0	1	0	1	0	1	0	0	0	0	1	0	1	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0
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45	0	1	0	1	1	0	1	0	1	1	0	0	1	0	1	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0
46	0	1	0	1	1	1	0	0	1	1	1	1	0	0	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0
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52	0	1	1	0	1	0	0	1	1	1	1	1	0	1	1	0	1	1	1	1	0	0	0	0	1	1	1	0	1	0	0
53	0	1	1	0	1	0	1	0	0	1	0	0	1	0	0	1	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0
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55	0	1	1	0	1	1	1	0	0	0	0	0	1	0	1	1	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0
56	0	1	1	1	0	0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1
57	0	1	1	1	0	0	1	0	0	0	1	1	1	0	1	0	0	0	1	1	0	0	1	0	1	1	1	0	1	0	1
58	0	1	1	1	0	1	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0	0	1	1	0	1	1	0	1	0	1

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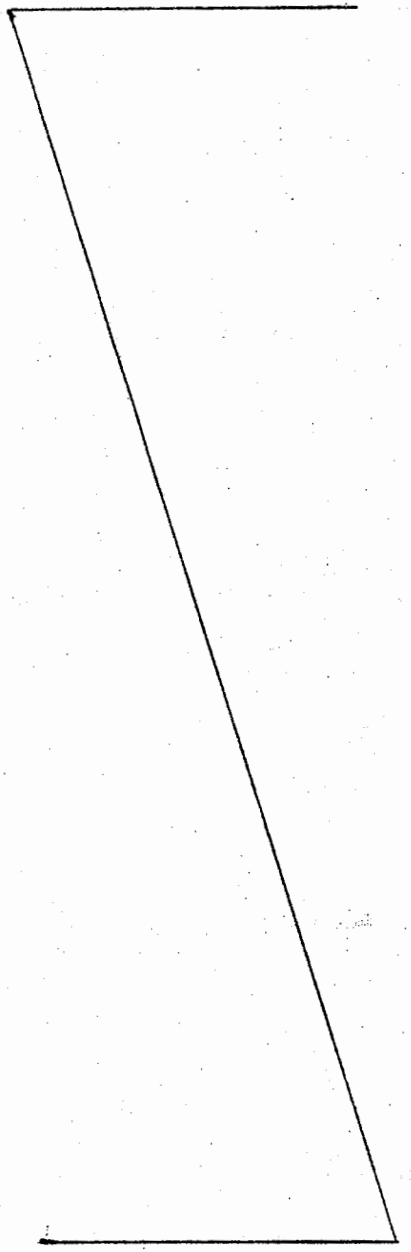
TABLE I - INPUT/OUTPUT MEMORY I TEST

	INPUT							OUTPUT																							
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59	0	1	1	1	0	1	1	0	0	0	0	0	0	0	0	1	1	0	1	1	0	1	1	1	1	0	0	1	0	1	1
60	0	1	1	1	1	0	0	0	0	0	1	1	1	0	0	0	0	1	0	1	0	1	1	0	0	0	1	0	0	1	1
61	0	1	1	1	1	0	1	0	0	0	0	0	0	0	0	1	0	1	1	1	0	1	1	0	1	0	1	0	0	1	0
62	0	1	1	1	1	1	0	0	0	0	0	0	1	1	0	1	1	1	0	1	0	0	0	0	0	1	1	0	0	0	0
63	0	1	1	1	1	1	1	0	0	0	1	0	1	0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	0	0
64	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	1	0	1	0	1	1								
65	1	0	0	0	0	0	1	0	0	0	1	1	1	0	1	0	0	0	0	1	0	1	1								
66	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	1	0	1	1	0	1	1								
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68	1	0	0	0	1	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0	1	1								
69	1	0	0	0	1	0	1	1	1	1	0	0	1	1	0	1	0	0	0	1	0	1	1								
70	1	0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0	1	0	1	0	1	1								
71	1	0	0	0	1	1	1	0	0	0	0	0	0	1	1	1	1	0	1	1	0	1	1								
72	1	0	0	1	0	0	0	0	1	0	0	0	1	0	0	1	0	1	1	1	0	0	0								
73	1	0	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	1	1	1	0	1	1								
74	1	0	0	1	0	1	0	1	1	1	0	0	1	0	1	1	0	0	0	1	0	1	1								
75	1	0	0	1	0	1	1	0	0	0	1	1	1	1	0	0	0	1	0	1	0	1	1								
76	1	0	0	1	1	0	0	0	0	0	1	1	1	1	0	0	0	1	0	1	0	1	1								
77	1	0	0	1	1	0	1	0	0	0	0	0	0	0	1	1	0	1	1	1	0	1	1								

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Dec.	INPUT							OUTPUT																				
	3	9	8	7	6	5	4	36	37	35	34	31	32	33	38	11	10	12	13	16	15	14	17	21	20	22	23	26	25	24	28
78	1	0	0	1	1	1	0	0	0	0	0	0	0	1	0	1	1	0	1	1	0	1	1								
79	1	0	0	1	1	1	1	0	0	0	1	1	1	0	1	0	0	1	0	1	1	0	0								
80	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	0	1	1								
81	1	0	1	0	0	0	1	0	0	0	0	0	0	0	1	1	1	0	1	1	0	1	1								
82	1	0	1	0	0	1	0	1	1	1	0	0	0	0	1	1	0	0	0	1	0	1	1								
83	1	0	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	0	1	1	0	1	1								
84	1	0	1	0	1	0	0	0	0	0	0	0	0	1	0	1	0	1	1	1	0	1	1								
85	1	0	1	0	1	0	1	0	0	0	0	0	0	0	0	1	0	1	1	1	0	1	1								
86	1	0	1	0	1	1	0	0	0	0	1	1	1	0	1	0	0	1	0	1	0	1	1								
87	1	0	1	0	1	1	1	0	0	0	0	0	0	0	0	1	0	1	1	1	0	1	1								
88	1	0	1	1	0	0	0	1	1	1	0	0	1	0	0	1	0	0	0	1	0	1	1								
89	1	0	1	1	0	0	1	0	0	0	0	0	0	0	0	1	1	0	1	1	0	1	0								
90	1	0	1	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	1	1	0	1	0								
91	1	0	1	1	0	1	1	0	0	0	1	1	1	0	0	0	1	0	0	1	0	1	0								
92	1	0	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	0	1	1	0	1	0								
93	1	0	1	1	1	0	1	0	0	0	1	1	1	0	0	0	0	1	0	1	0	1	0								
94	1	0	1	1	1	1	0	1	0	0	1	1	0	1	0	0	1	1	1	1	0	1	0								
95	1	0	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	0	1	1	0	1	0								
96	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	1	0	1	0	1	0								
97	1	1	0	0	0	0	1	0	0	0	0	0	0	1	1	1	1	0	1	1	0	1	0								



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TABLE I - INPUT/OUTPUT MEMORY I TEST

	INPUT							OUTPUT																			
Dec.	3	9	8	7	6	5	4	36	37	35	34	31	32	33	38	11	10	12	13	16	15	14	17	21	20	22	23	26	25	24	28
98	1	1	0	0	0	1	0	0	0	0	1	1	1	1	1	0	1	0	0	1	0	1	0								
99	1	1	0	0	0	1	1	0	0	0	0	0	0	1	0	1	0	1	1	1	0	1	1								

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4.2.1.3 Digital voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply.- All the requirements shall be met with a power supply of + 5.00 ± 0.05 VDC.

4.2.2.2 Digital output.- The "0" or false-output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between + 2.40 and 5.00 VDC.

4.2.2.3 Test set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.4 Input/Output Test.- Connect the Circuit Card as shown in Figure I. When digital "1"s, and "0"s, as defined in paragraph 4.2.2.2 above, are applied to the respective input pins on Table 1 the corresponding output shall be as shown.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11745203.

6. NOTES

6.1 The notes specified in the general specification MIL-G-48411 are applicable to this specification.

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